Ashutosh Tiwari

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Work Experience

Senior Machine Learning Engineer (ML Platform & Frameworks) — Jul. 2024 – Present
ADOBE FIREFLY — *GENERATIVE AI, COMPUTER VISION, LARGE LANGUAGE MODELS*

- Lead fault-tolerant distributed inference and training infrastructure for the org's ML platform: GPU pipelines on Kubernetes engineered to survive node loss and partial failure, from 40-node daily EMR ingestion through production model inference.
- **Distributed Inference Frameworks**: Designed and led engineering for the org's online and offline inference systems powering experimentation and production at scale, reaching 24×A100-40GB per model job across 9 model queues and 64 H100s in one parallel wave.
  - Built online inference on vLLM + Ray for rapid experimentation with LLMs such as Falcon-40B and Llama 2 70B.
  - Built offline inference on Ray Data + PyTorch Lightning; a module benchmarked on that engine at 1.09M rows, 8×A100, 16 fractional-GPU actors. Introduced torch.compile(mode="max-autotune") in production: compiler-generated Triton kernels, fusion, CUDA-graph capture.
  - Modeled pipelines as horizontally-scaling Source → Transform → Sink plugins with no pod-to-pod communication, KEDA-autoscaled on SQS queue depth; batched sends retry on exponential backoff with full jitter, and undeliverable work drains to a DLQ.
- **Foundation Model Training Framework**: Leading development of a PyTorch-based training framework adopted across the org for training generative models on billions of images and videos.
  - Designed a plugin-based strategy pattern over FSDP/FSDP2 to make distributed training strategies pluggable.
  - Integrated Flash Attention 2/3, context parallelism, and distributed attention for DiT text-to-image and text-to-video training.
- **Build & Deployment System**: Built a CLI-driven build system (BuildKit images on KEDA-scaled Kubernetes jobs, pushed to ECR), fronted by a Rust control plane and an MCP interface for AI-agent-driven builds, deployments, and inference.
- **Enrichment Service**: Orchestration over the inference framework; migrated its 202.6M-row and 24.8M-clip datasets on idempotent re-runs; ran fire-and-forget pipelines behind a concurrency-safe S3 checkpoint cache, lease/TTL heartbeats and a circuit breaker.
- **Training-Data Governance**: Solely authored the org's fault-tolerant training-data lineage subsystem in Rust (27,135 lines): idempotent claim/receipt registration, immutable S3 evidence layouts, and manifest-after-data publication, proven by a 9.7k-line integration test.
- **Data Quality Framework**: Wrote the org's first data quality framework on Cerberus, used across enrichment pipelines to validate the correctness of feature-generation outputs across a ~700M-asset catalog growing 35–38M images/month.

Software Dev Engineer II (ML Platform) — Jan. 2019 – Jul. 2021
SWIGGY — *DATA SCIENCE PLATFORM, TIME SERIES FORECASTING*

- **Online Inference Platform**: Core contributor to the platform that served real-time model inference to 18M+ monthly transacting users across food delivery and quick commerce, autoscaled on live load metrics.
  - Engineered the dynamic request-routing layer that matched incoming API calls against geo-tags, traffic profile and service metadata, then dispatched them to model clusters partitioned by region and load domain.
  - Implemented routing and dispatch on an Akka-style actor framework: resilient asynchronous invocation, fault isolation across clusters, and backpressure-aware scheduling that held sustained low P99 under high QPS.
- **Feature Store**: Built and maintained the feature store and its pipelines, serving on-demand features to production ML models at 4Bn rows and 10K QPS, with multichannel ingestion via Spark, Flink, and user files.
- **Forecasting & Correlation Platform**: Founding member of the time-series forecasting platform adopted by teams across the org; its forecasts drove critical scaling decisions in real time.
- **DAQ**: Led DAQ, a large-scale API scraping tool collecting 15M rows daily for analysis and model training.

Software Development Engineer (Search Relevance) — Sep. 2017 – Jan. 2019
FLIPKART (a Walmart company) — *SEARCH RELEVANCE, QUERY INTENT, NLP*

- **Search Intent Models**: Built and improved CRF and neural-network intent models powering search and discovery for millions every day; drove the error-class analysis, designed the fixes and shipped them.
- **Tail Query Classifier**: Shipped a FastText-based query-store classifier that predicts the category of a tail query.
- **ML Workflow Automation**: Built the first workflow at Flipkart to retire manual deploys, automating training and auto-deployment of search models—written first in Luigi and later migrated to Airflow.
  - Wrote a generic Airflow framework that builds DAGs at runtime per ML model and orchestrates the train→deploy flow, gating every auto-deploy on data and model validations.
- **Large-Scale Data Pipelines**: Implemented Cascading/HDFS pipelines over 4Bn+ user-event datapoints, extracting and transforming them into the training data the search models learned from.

Software Development Engineer — Sep. 2016 – Sep. 2017
GROUPON — *BACKEND ENGINEERING*

- *Customer segmentation and deal-recommendation ML (NSVD unsupervised collaborative filtering on Neo4j); Cyclops customer-service platform live in every Groupon country.*

Software Engineer — Sep. 2015 – Aug. 2016
NETSPEED SYSTEMS (Acquired by Intel) — *GRAPH ALGORITHMS, NETWORK ON CHIP*

- *C++ graph algorithms for Network-on-Chip interconnects: Virtual Channel Arbitration, Polarity-based Arbitration, Multi-Cast Filtering, Structural Latency Breakdown.*

Publications

Accepted at *NetSci 2023* (Poster Presentation) and *IC2S2 2023* (Parallel Talk) as **first author**

[1] **Ashutosh Tiwari**, Prof. Sadamori Kojaku, Prof. Yong-Yeol Ahn, "Biased Contrastive Learning debiases Graph Neural Networks," *International Conference on Network Science (NetSci)*, 2023. Poster

In this work, we propose a non-parametric contrastive learning framework to learn debiased graph embeddings with respect to sensitive node attributes and structural homophily. Through empirical evaluations on different datasets, we demonstrate that our method offers a better approach to debiasing compared to existing approaches and thus results in more organic recommendations across different GNN architectures.

Education

Indiana University, Bloomington
Master of Science in Computational Data Science 3.87/4.0 — *Aug. 2021 – May 2023*

National Institute of Technology
Bachelor of Science in Computer Science 8.32/10.0 — *Aug. 2011 – May 2015*

Research Experience / Independent Study

- Implementing novel training frameworks that result in unbiased (or fair) recommendation systems from "biased" datasets at IUNI with Prof. YY Ahn and Prof. S Kojaku as part of my Independent Study. — Summer 2022 - Summer 2023
- Working as a paid RA for, **"User Intent as a Network"** with Prof. YY Ahn, P Kantak and FB Yara. Project is funded by Kelly Business School. Goal is to be able to quantify and thus act upon that "intent" network. — Summer 2022 - Fall 2022
- Was part of NLP Lab@IUB with Prof. D Cavar. Contributed to design of TieML and Events' Timeline modelling using different fine-tuned Large Language Models (LLMs). — Fall 2021

Teaching Experience

- Teaching Assistant for Network Science (INFO-I 606) with Prof. YY Ahn — Spring 2023
- Teaching Assistant for Machine Learning (CSCI-B 555) with Prof. R Khardon — Fall 2022
- Teaching Assistant for Network Science (INFO-I 606) with Prof. YY Ahn — Spring 2022

Selected Projects

BiasNet | *Deep Reinforcement Learning, PyTorch, Actor Critic Algorithm, On-policy Model Free* — Code/Report
- Learning to fight in Street Fighter II with induced relational bias from differential game scenes.

DeepFoodie | *Python, Tensorflow, Self Supervised Deep Clustering, Deep Learning, Transfer Learning* — Code/Report
- Clustering dishes on basis of their ingredient embeddings. These ingredient embeddings are generated by a NN.

BlindNet | *Python, Pytorch, Deep Learning, Transfer Learning* — Code/Report
- Image to vector generation on Coco dataset.

Continuous Dominant Set Repair | *C++, Graph Algorithms, Guha and Khuller's Algo., Greedy* — Code
- Repairs a broken link in Continuous Dominant Set in $O(\Delta^2)$, where Δ being the avg cardinality of connected graph.

Humana Mays Healthcare Analytics Case Competition | *Boosted Trees, Feature Engineering* — Code
- 11th rank on leaderboard. Hosted by TAMU and Humana Mays, 2021.

AnalyticsVidhya Solutions | *Python, Tensorflow, Torch, Time Series, Feature Engineering, Catboost* — Code

Investigating Bias Manifolds | *Bias Manifolds, Bias Progression, Measuring Bias, Python, Word2vec* — Code/Report

Technical Skills

Search Relevance, Recommendations, Graph Neural Networks, Fairness Aware Modeling, Computational ML, NLP, Computer Vision, DL, Spark, Python, Scala, C++, Contrastive Learning, Large Language Models
Frameworks/Libraries/Tools: Ray, Pytorch Geometric, Pytorch Lightning, Pytorch, Tensorflow, Sklearn, Numpy, Pandas, Matplotlib, HDFS, Spark, Kafka, Flink, AWS Sagemaker, DynamoDB, Faiss, vLLM, Django